

# **Collab-flow Project**

## **Final Report**

### **Project Tittle: Collab-flow**

**Generation: 11**

**Group: 8**

#### **Team Member:**

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# Table of Contents

1. Introduction

2. User Research & Problem Definition

3. Ideation & Conceptual Design

4. Prototyping & Iterative Design

- 4.1. Information Architecture & User Flow
- 4.2. Functional & Non-Functional Requirements
- 4.3. Design Principles & Heuristic Evaluation
- 4.4. Usability Testing & Evaluation
- 4.5. Iteration & Final Design Refinement

5. Conclusion

# 1. Introduction

## Project Overview

Collab-Flow is a comprehensive project collaboration system designed specifically for university students who struggle with managing group projects. The platform provides an all-in-one solution that combines task management, real-time communication, AI assistance, and team coordination tools.

## Problem Statement

University students often face significant challenges when working on group projects, including:

- Difficulty managing time across multiple projects
- Unclear task ownership and responsibilities
- Fragmented communication across multiple platforms
- Overwhelming cognitive load from complex tasks
- Unequal participation and free-riding
- Difficulty getting started with ideation and planning

## HCI Design Process Overview

We followed a complete HCI design lifecycle from Week 2 to Week 9, applying user-centered design principles at every stage:

| Phase         | Weeks | Activities                                                                                                        |
|---------------|-------|-------------------------------------------------------------------------------------------------------------------|
| User Research | 2-3   | Task analysis, personas, user journeys, accessibility audit                                                       |
| Ideation      | 4-5   | HMW questions, brainstorming, storyboarding, design critique                                                      |
| Prototyping   | 6-9   | Information architecture, wireframes, high-fidelity prototype, heuristic evaluation, usability testing, iteration |

## **2. User Research & Problem Definition**

### **2.1. Understanding Users and Stakeholders**

The main users of Collab-Flow are university students who have project work but struggle with managing their time and collaborating effectively.

**Stakeholders include:**

- Project Manager (Primary Users)
- Team Member / Developer
- Designer / Content Creators
- Stakeholders / Clients
- System Administrator

### **2.2. Task Analysis**

**Key User Tasks Identified:**

- Register new account / Login to existing account
- Create new project
- View project dashboard
- Create and assign tasks
- Update task status (To Do → In Progress → Review → Done)
- Add comments to tasks
- Invite team members
- Receive and respond to notifications

### **2.3. Context of Use**

Users interact with the system in the following contexts:

- During study hours (9 AM - 5 PM)
- On desktop/laptop computers
- Sometimes on mobile devices (remote work)
- Under deadline pressure
- While multitasking between projects

**Key Findings from User Research:**

- Users prefer fast and simple task management

- Repetitive task creation is inconvenient
- Users often work on multiple projects simultaneously
- Status updates need to be one-click actions
- Notifications are critical but easily become noise

## 3. Ideation & Conceptual Design

### 3.1. Ideation Approach

We used brainstorming techniques to explore multiple solutions and then refined them into an integrated project collaboration system.

#### How Might We (HMW) Questions:

| Type                    | Question                                                                                                                         |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Assumption-Based</b> | "How might we redesign group work so students can collaborate effectively even with different working styles?"                   |
| <b>Analogy-Based</b>    | "How might we make group project collaboration feel as coordinated and role-driven as playing Mobile Legends: Bang Bang (MLBB)?" |
| <b>Emotion-Based</b>    | "How might we help students feel more organized, confident, and aware of their responsibilities during group projects?"          |

### 3.2. Selected Solution Idea

We designed a comprehensive project collaboration system (Collab-Flow) where users can:

- Manage multiple projects in one dashboard
- Track tasks with visual status indicators
- Receive intelligent notifications
- Collaborate in real-time with team members

### 3.3. Storyboarding

#### Scenario 1: Developer David updates task status

- 1. David logs into Collab-Flow
- 2. Dashboard shows "My Tasks" section
- 3. Sees assigned task: "Fix login bug"
- 4. Clicks "Mark In Progress" (one click)
- 5. Task moves to In Progress column
- 6. Project Manager Priya receives real-time notification
- 7. Team sees updated progress on dashboard

#### Scenario 2: New user Nina creates first project

- 1. Nina completes registration
- 2. Guided tour starts automatically
- 3. Tooltip shows "Click here to create your first project"
- 4. Nina fills project name and description
- 5. Invites team members via email
- 6. Dashboard populates with sample tasks
- 7. Nina feels confident to continue

### 3.4. Initial Design Critique & Feedback

| Feedback                               | Action Taken                             |
|----------------------------------------|------------------------------------------|
| Dashboard feels cluttered              | Reorganized into 3-column layout         |
| Color alone for status is inaccessible | Added patterns + text labels             |
| Notification panel needs grouping      | Grouped by project with expand/collapse  |
| Mobile navigation confusing            | Added bottom navigation bar              |
| Invite member button hard to find      | Moved to prominent position, added color |

## **4. Phase 3: Prototyping & Iterative Design**

### **4.1. Information Architecture & User Flow**

#### **User Flow: Create New Project**

Start

Click "+ New Project"

Fill Project Details (name, description, deadline, priority)

Invite Team Members (emails, assign roles)

Review & Confirm

Project Created

Redirect to Project Board

End

#### **Main Navigation Structure:**

| Section              | Content                                                                 |
|----------------------|-------------------------------------------------------------------------|
| <b>Dashboard</b>     | Project summary cards, My Tasks, Recent activity feed                   |
| <b>Projects</b>      | All Projects, Create New Project, Project Details (Tasks, Team Members) |
| <b>My Tasks</b>      | Assigned to Me, Task Details (Comments, Attachments)                    |
| <b>Notifications</b> | Unread, All Notifications                                               |
| <b>Team</b>          | My Teams, Team Details                                                  |
| <b>Profile</b>       | Account Settings, Preferences, Logout                                   |

## Navigation Structure:

- Desktop: Sidebar navigation + Top header
- Mobile: Bottom navigation bar (5 tabs)
- Step progress indicator for multi-step processes
- Breadcrumbs for deep navigation
- **Confirmation modal after destructive actions**

## 4.2. Functional & Non-Functional Requirements

### Functional Requirements:

| Module                | Functions                                             |
|-----------------------|-------------------------------------------------------|
| <b>Authentication</b> | Register, Login, Logout, Forgot password, JWT refresh |
| <b>Projects</b>       | Create, Read, Update, Delete, Archive, Search         |
| <b>Tasks</b>          | Create, Assign, Update status, Comment, Attach files  |
| <b>Team</b>           | Invite members, Assign roles, Remove members          |



| Module                | Functions                                                                                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Notifications         | Real-time alerts, Email digest, Mark as read                                                                                                       |
| Dashboard             | Widgets, Charts, Activity feed                                                                                                                     |
| <b>+ NEW FEATURES</b> |                                                                                                                                                    |
| Module                | New Functions                                                                                                                                      |
| AI Assistance         | AI project ideation, AI task breakdown, AI study guide generator, AI document summarization, AI writing assistant, AI deadline manager             |
| Invitation & Chat     | Project-based chat, Direct messaging, Invite by email, Invite by QR code, Invite by share link, @mentions, Pinned messages, Role-based permissions |
| Subtasks              | Nested subtasks, Independent status per subtask, Individual assignees, Subtask due dates, Progress tracking, Bulk operations                       |
| Participation Monitor | Contribution tracking, Activity analytics, Participation alerts, Peer recognition, Anonymous feedback                                              |

## Non-Functional Requirements:

| Requirement   | Specification                                  |
|---------------|------------------------------------------------|
| Performance   | Page load < 2 seconds, API response < 200ms    |
| Usability     | Simple and clean UI, < 3 clicks for core tasks |
| Accessibility | WCAG 2.1 AA compliant, keyboard navigation     |
| Reliability   | 99.9% uptime, real-time updates < 500ms        |

|             |                                                     |
|-------------|-----------------------------------------------------|
| Requirement | Specification                                       |
| Security    | JWT authentication, password hashing, rate limiting |
| Scalability | Support 1000+ concurrent users                      |
| Mobile      | Responsive design, PWA capable                      |

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### 4.3. Design Principles & Heuristic Evaluation

We want to critically evaluate the Collab-Flow prototype against established usability heuristics and apply design principles to improve the user experience.

#### Affordances & Signifiers

| Interface Element    | Affordance                        | Signifier                                | Improvement Made                                      |
|----------------------|-----------------------------------|------------------------------------------|-------------------------------------------------------|
| Task Cards           | Drag to reorder or change status  | Initially only dropdown menu             | Added handle icon (:::) and "grab" cursor on hover    |
| "Create Task" Button | Click to create a new task        | Standard blue button                     | Changed to Floating Action Button (FAB) with "+" icon |
| Project Name         | Click to navigate to project home | Underlined on hover                      | Added home icon and permanent underline               |
| AI Assistant Icon    | Click to activate AI help         | Sparkle icon with subtle pulse animation | Clear visual differentiator from other buttons        |
| Chat Icon            | Click to open project chat        | Speech bubble icon with unread badge     | Real-time notification count                          |

## *Feedback, Constraints, and Mapping*

### **Feedback:**

- Smooth animation when task card moves to new column
- AI response shows typing indicator while generating content

### **Constraints:**

- Project Name field: Required, minimum 3 characters
- Inline validation with error messages
- Confirmation modal before deleting projects or tasks

### **Mapping:**

- Kanban board columns: To Do → In Progress → Review → Done (natural workflow mapping)
- Chat messages: Newest messages at bottom (natural conversation mapping)

## **4.4. Usability Testing & Evaluation**

We want to evaluate the improved prototype through formal usability testing with real users and translate findings into actionable design changes.

### **Why We Test**

We conducted usability testing to:

- Identify where users get stuck (efficiency bottlenecks)
- Discover what users misunderstand (conceptual problems)
- Measure user satisfaction (SUS score)
- Validate our new AI, chat, and subtasks features

### **Think-Aloud Method**

We used the Think-Aloud method during user testing. Participants verbalized their thoughts, feelings, and reasoning as they performed tasks.

### **Example Think-Aloud Session Notes:**

*"Okay, I'm on the dashboard. I want to create a project... I see the 'New Project' button... I click it. Oh, this is a wizard! Step 1: Details, Step 2: Invite Members, Step 3: Review. That's much better than before. I'm filling in the details... Now Step 2, I can invite members... I'll paste their emails... Done. Wow, that was easy!"*

## 4.5. Iteration & Final Design Refinement

We want to synthesize all learning from the HCI process, create the final design rationale, and reflect on team collaboration and growth.

### Synthesis of Learning Across the Course

| Week     | Focus                      | Key Takeaway                                                          |
|----------|----------------------------|-----------------------------------------------------------------------|
| Week 2   | User Research              | Understanding users' pain points is the foundation of any good design |
| Week 3   | Personas & Accessibility   | Designing for "extreme users" leads to better designs for all users   |
| Week 4-5 | Ideation & Storyboarding   | Creative methods help us think outside the box                        |
| Week 6   | Information Architecture   | A well-structured system is the backbone of good UX                   |
| Week 7   | Heuristics & Principles    | Established guidelines help systematically evaluate designs           |
| Week 8   | Usability Testing          | Watching real users is the most valuable feedback                     |
| Week 9   | Iteration & AI Integration | AI can significantly enhance collaboration and reduce cognitive load  |

#### Final Design Rationale

When presenting Collab-Flow, we will persuade stakeholders with these arguments:

1. **User-Centered from the Start:** We began with in-depth user research, creating personas (Priya, David, Maria) that represent our target audience. Every design decision was informed by their needs.
2. **Data-Driven Iteration:** We tested with real users and made changes based on quantitative data (SUS scores, task completion rates) and qualitative data (think-aloud quotes).
3. **Solve a Real Problem:** Collab-Flow directly addresses the pain points of university students: messy communication, unclear task ownership, and lack of a unified platform.
4. **AI-Powered Innovation:** Our AI features (ideation, study guides, summarization, participation monitoring) set Collab-Flow apart from traditional project management tools and specifically address student needs.

## Final Solution Impact

The final Collab-Flow solution improves project collaboration by:

- **Reducing repetitive task management** through templates and quick actions
- **Eliminating context switching** with an all-in-one platform
- **Improving accessibility** with WCAG 2.1 AA compliance
- **Enhancing real-time collaboration** with instant updates and notifications
- **Simplifying team coordination** with role-based permissions
- **Accelerating project planning** with AI-powered ideation and task breakdown
- **Centralizing communication** with built-in chat and invitation tools
- **Making large tasks manageable** with subtasks and progress tracking
- **Promoting fair participation** with contribution monitoring

## 5. Conclusion

Reflection on the HCI Journey

Throughout the 10-week journey, we learned that great design is not a linear process but a continuous cycle of **research -> ideate -> prototype -> test -> iterate**.

## What We Learned About Collaboration

- **Communication:** Regular meetings and clear division of responsibilities were key to our team's success.
- **Respect for the Process:** We learned to trust the HCI design process. Even when we felt we had a "good enough" solution, user testing always revealed something we could improve.

## What We Would Do Differently

- Start usability testing earlier (Week 6 instead of Week 8) to allow more time for iteration
- Implement more AI features earlier to test their impact on collaboration

## Personal Reflections

| Team Member         | Reflection                                                                                                                                                                                                                                            |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Nob Johnson</b>  | "I learned that as a developer, I tend to focus on functionality first. This course taught me to prioritize the user's experience above all else. It's not enough for a feature to work—it has to be intuitive and delightful to use."                |
| <b>Long Kimneng</b> | "The most impactful lesson for me was the importance of accessibility. I had never considered how a user with a screen reader would interact with a website. Now, it's a core part of my design thinking."                                            |
| <b>Kol Udom</b>     | "I used to think design was all about aesthetics. Now I understand it's a strategic problem-solving process. Heuristics, user journeys, and usability testing are tools for making design decisions that are both creative and grounded in evidence." |

## Future Enhancements

Based on our research and user feedback, we envision the following future features:

1. **Voice-to-Text Task Creation:** Students can speak to create tasks and subtasks

2. **AI Meeting Summary:** Automatically generates meeting notes and action items from chat
3. **AI Conflict Mediator:** Helps resolve team conflicts by suggesting compromises
4. **Gamification:** Points, badges, and leaderboards for completed tasks
5. **Integration with Google Workspace & Microsoft 365:** Seamless document editing
6. **AI-Powered Peer Evaluation:** AI-assisted peer evaluation forms with sentiment analysis

## Final Summary

The Collab-Flow system was developed through a complete HCI design process:

- **Week 2:** Understanding users and defining the problem - Task analysis, user research
- **Week 3:** Creating personas and user journeys - 4 personas, journey maps, accessibility audit
- **Week 4-5:** Ideation and selecting collaboration solution - Storyboards, design critique, SCAMPER
- **Week 6:** Building prototype and defining system requirements - Information architecture, functional specs, database design
- **Week 7:** Design principles and heuristic evaluation - Affordances, signifiers, Nielsen's 10 heuristics
- **Week 8:** Usability testing and evaluation - Think-aloud method, test planning, data analysis
- **Week 9-10:** Iteration and final refinement - AI integration, new features, team reflection

**The final solution improves project collaboration by:**

- Reducing cognitive load through AI assistance
- Eliminating context switching with an all-in-one platform
- Improving accessibility with WCAG 2.1 AA compliance
- Enhancing real-time collaboration with instant updates and notifications
- Simplifying team coordination with role-based permissions
- Promoting fair participation with contribution monitoring